

YOUNME DATING APP

MASTER TECHNICAL ARCHITECTURE SPECIFICATION & SYSTEM SPEC (v3.2)

Production Runtime: GCP Cloud Run Tri-Microservices (asia-south1 Mumbai) | **Stack:** Go Engine • NestJS • Cloud Run Worker (BullMQ) | **Data & Media:** Supabase Pro (PostgreSQL + PostGIS) • Memorystore Redis • LiveKit Cloud

1. EXECUTIVE SUMMARY & CORE ARCHITECTURE DRIVERS

YoUnMe is a high-concurrency, mobile-first social discovery platform engineered on a containerized microservices topology. The architecture unifies gesture-based swipe matching, real-time presence tracking, low-latency WebRTC audio/video calls, and a secure creator coin economy. Production workloads are orchestrated on Google Cloud Run in the asia-south1 (Mumbai) region across three decoupled services: a Node.js/NestJS API service, a high-throughput Go real-time engine, and a dedicated Cloud Run background worker consuming asynchronous tasks via BullMQ from GCP Memorystore for Redis. Relational persistence and geospatial indexing are governed by Supabase Pro (PostgreSQL + PostGIS), while external communications leverage Twilio WhatsApp template OTP with automated Firebase Auth SMS failover, LiveKit Cloud for WebRTC media sessions, and Google Cloud Vision API for automated media safety.

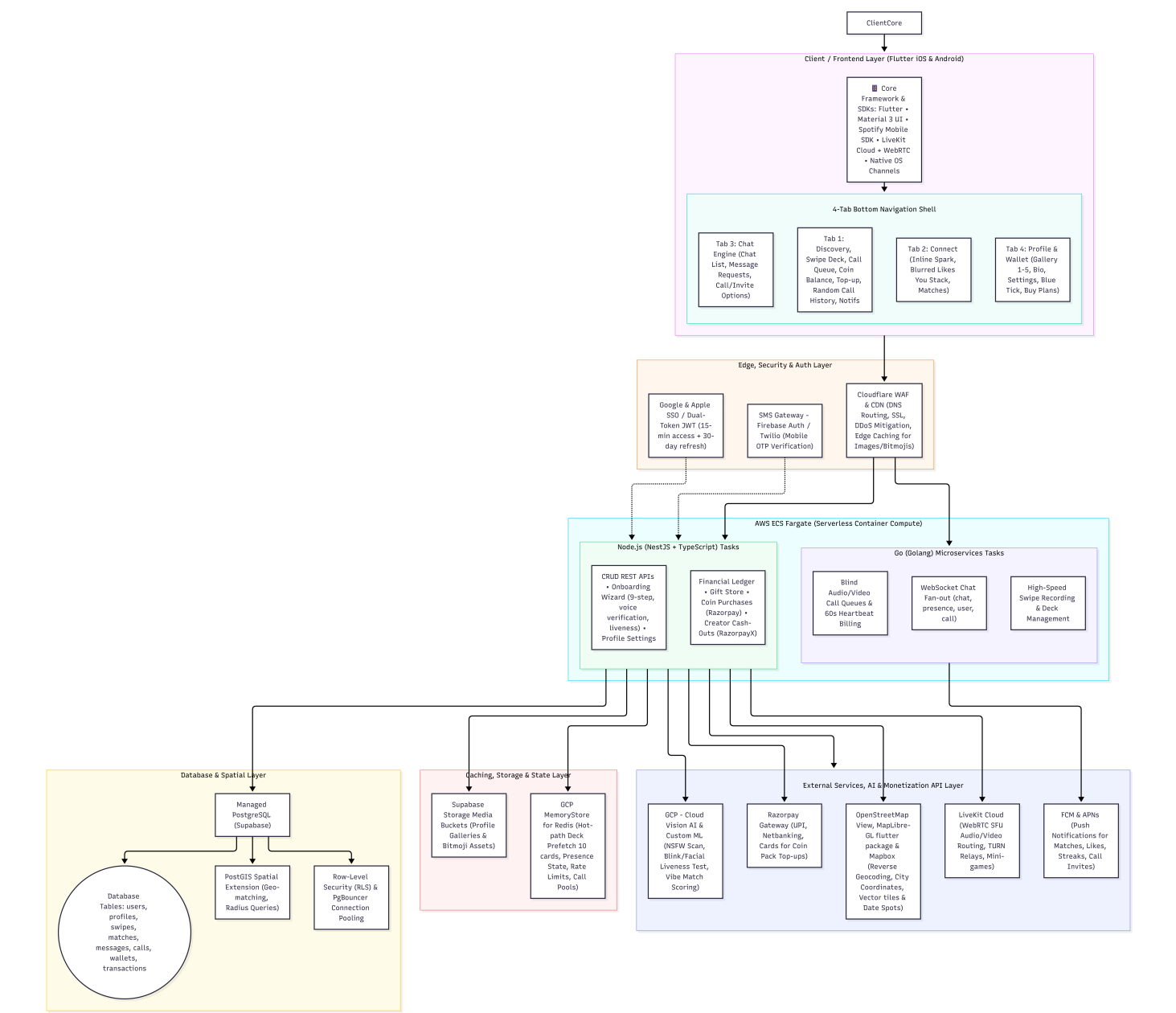
Attribute	Target Driver & Architectural Requirement	Technical Mechanism & Finalized Stack
Compute Scalability	Sub-second horizontal autoscaling; decoupled execution for stateless APIs, WebSocket events, and async background workers.	Tri-service Google Cloud Run microservices in asia-south1 (Mumbai): NestJS API (0.25 vCPU, 512 MiB, 20 req/inst), Go Engine (1 vCPU, 1 GiB, 100 req/inst), and Background Worker (1 vCPU, 512 MiB, auto-scales to zero).
Trust & Safety	Automated multi-tier content moderation, biometric liveness verification, and in-flight communication guardrails.	Google Cloud Vision API (Image Text Detection & Safe Search explicit content filtering), Blink Liveness SDK, and Live Secure Chat Link Guard.
Low-Latency State & Queues	API p95 < 300 ms; swipe refill < 200 ms; call signaling < 1 s; real-time presence and task queue orchestration.	GCP Memorystore for Redis (Basic Tier, 1 GiB) for hot-path candidate deck prefetching (10 cards), presence heartbeats, and BullMQ task queue management.
Real-Time Media	Low-latency WebRTC peer-to-peer and SFU audio/video calls, blind matchmaking, and in-call interactive experiences.	LiveKit Cloud (Ship Plan) powering WebRTC peer-to-peer media sessions, dynamic room lifecycle management, and SFU selective forwarding.
Relational & Media Storage	High-integrity persistence, zero double-spending, PostGIS geo-discovery supporting up to 100k MAU, and multi-asset storage.	Supabase Pro (PostgreSQL + PostGIS with 8 GB DB storage, 100 GB storage bucket, supporting 6 user photos + 1 appearance avatar per user, anonymous call avatars, RLS) + double-entry ledger.
Auth & Security Messaging	High-delivery OTP verification, multi-channel automated fallback, enterprise single sign-on.	Primary Twilio WhatsApp template OTP with automated fallback to Firebase Auth Phone Auth / SMS; Google SSO and Apple SSO via Supabase Auth.
Observability & Logging	Centralized telemetry, latency tracing, system health checks, and resource saturation alerting.	GCP Cloud Logging (50 GiB log storage, 1-month retention) and Cloud Monitoring for real-time telemetry, error tracking, and instance metrics.

2. MASTER TECHNOLOGY STACK ARCHITECTURE

YoUnMe adopts a modern cloud-native architecture decomposed into distinct client, security perimeter, compute, data, and external service tiers. The complete end-to-end cloud topology and component workflow is illustrated in Figure 1 on the following page:

2. MASTER TECHNOLOGY STACK ARCHITECTURE (CONTINUED)

Figure 1: YoUnMe Cloud-Based Web & Mobile Architecture & End-to-End Workflow Diagram



Architectural Workflow Summary: The end-to-end user journey starts at the 4-tab Flutter mobile client (Discovery, Connect, Chat Engine, Profile & Wallet). Client requests traverse Cloudflare's security perimeter (DNS, SSL/TLS, DDoS mitigation, and edge CDN) to Google Cloud Load Balancing (GCLB). GCLB routes synchronous REST traffic to the NestJS API service and persistent WebSocket streams (/ws/*) to the Go Real-Time engine on GCP Cloud Run (asia-south1). Asynchronous workloads (Vision AI moderation, batch push notifications) are enqueued into GCP Memorystore for Redis and processed by the dedicated Cloud Run Background Worker via BullMQ. Data persistence, spatial matching (PostGIS), and media assets are governed by Supabase Pro, while live audio/video calls are routed via LiveKit Cloud SFU.

2. MASTER TECHNOLOGY STACK ARCHITECTURE (CONTINUED)

Detailed Technical Breakdown of YoUnMe MVP Production Stack

The table below details each architectural layer, the finalized technology choice, and the specific engineering responsibilities and runtime configurations:

Architectural Layer	Finalized Technology	Engineering Scope, Configuration & Responsibilities
Client / Frontend	Flutter (iOS & Android) MapLibre GL Mobile	Single cross-platform codebase, Material 3 UI shell, 5-tab navigation. Integrates MapLibre GL open-source map rendering with Mapbox/OSM vector tiles, LiveKit WebRTC SDK, and FCM/APNs push listeners.
Core API Backend	Node.js (NestJS + TypeScript) GCP Cloud Run (asia-south1)	Primary RESTful API service configured with 0.25 vCPU, 512 MiB memory, 20 concurrent requests/instance. Executes user onboarding, profile settings, swipe logic, gift store, and financial ledger.
Real-Time Engine	Go (Golang) Microservice GCP Cloud Run (asia-south1)	High-concurrency event engine configured with 1 vCPU, 1 GiB memory, 100 concurrent requests/instance. Manages persistent WebSocket fan-out, user presence heartbeats, rapid swipe ingestion, and call matchmaking.
Background Worker	Cloud Run Background Worker asia-south1 + BullMQ	Dedicated asynchronous worker configured with 1 vCPU, 512 MiB memory, auto-scaling down to zero when idle. Consumes background tasks from Redis queue via BullMQ for Vision API scans and batch push dispatch.
Database & Spatial	Supabase Pro Managed PostgreSQL + PostGIS	Primary relational system of record (8 GB DB storage, 100 GB file storage, supporting up to 100k MAU). PostGIS spatial indexing (ST_DWithin) for geo-radius matching. Row-Level Security (RLS) enforcement.
Media & Asset Storage	Supabase Storage + Cloudflare CDN	Object storage supporting 6 profile photos + 1 appearance avatar per user, plus system reference strings for random anonymous call avatars, distributed through Cloudflare edge caching.
Cache & Queue State	GCP Memorystore for Redis (Basic Tier, 1 GiB)	Fast-access in-memory cache for session management, active user presence tracking, hot-path candidate deck prefetching (10 cards), rate limiting, and BullMQ message queue broker.
Networking & Perimeter	Global Cloud Load Balancing Cloud NAT & Cloudflare	GCLB with 1 forwarding rule as primary entry point. Cloud NAT (Public NAT, 2 VM instances, 1 IP address) for secure outbound traffic. Cloudflare Free Tier for DNS, SSL/TLS, and basic DDoS mitigation.
Container Registry	GCP Artifact Registry (Asia Repository)	Regional Docker container registry storing multi-stage production images, accelerating Cloud Run cold starts during horizontal autoscaling events.
Audio/Video Real-Time	LiveKit Cloud (Ship Plan)	Production WebRTC media routing powering peer-to-peer and SFU audio/video calls, room lifecycle management, automated token issuance, background blur, and media quality adaptation.
AI & Moderation	Google Cloud Vision API + Custom ML Models	Automated media safety: Image Text Detection (15k monthly requests) to intercept phone numbers/external handles, and Safe Search explicit content filtering (30k monthly requests). Dynamic Vibe Match ML.
Auth & Messaging	Twilio WhatsApp OTP Firebase Auth & Supabase SSO	Primary phone verification via Twilio WhatsApp template messages with automated fallback to Firebase Auth Phone Auth / SMS. Google SSO and Apple SSO integrated via Supabase Auth.
Push & Discovery	Firebase Cloud Messaging (FCM) APNs MapLibre GL	FCM for Android and APNs for iOS push notifications (matches, messages, calls, streaks). MapLibre GL combined with Mapbox / OpenStreetMap vector map tiles for interactive geospatial discovery.
Observability & Logging	GCP Cloud Logging & Cloud Monitoring	Centralized telemetry: GCP Cloud Logging (50 GiB storage, 1-month retention) and Cloud Monitoring tracking Cloud Run CPU/RAM saturation, worker queue depth, Redis memory, and latency metrics.

2. MASTER TECHNOLOGY STACK ARCHITECTURE (CONTINUED)

Cross-Platform Technology Stack Comparison (Tinder vs FRND vs Bumble vs YoUnMe)

A comprehensive engineering comparison across four modern social discovery and matchmaking platforms evaluates infrastructure trade-offs, architectural maturity, and technological agility. YoUnMe's finalized stack leverages cloud-managed serverless primitives to achieve enterprise-grade scale with minimal operational overhead:

Stack Layer	Tinder	FRND	Bumble	YoUnMe (Updated MVP Stack)
Mobile Frontend	Native (iOS / Android) (Swift, Kotlin)	Mobile App (Framework hidden)	Native (iOS / Android) (Swift, Kotlin)	Flutter (Single Codebase)
Frontend UI	Native UI	Not publicly disclosed	Native UI	Dart + Material 3 + MapLibre GL
Backend	Node.js, Java, Scala, Go (Microservices)	Not publicly disclosed	Node.js, JavaScript, PHP (Microservices)	Node.js + NestJS & Go (Tri-Microservices)
API / Gateway	Spring Cloud Gateway	Not publicly disclosed	Backend APIs / Internal Services	REST APIs + WebSockets (/ws/)
Database	PostgreSQL, Cassandra, DynamoDB	Not publicly disclosed	MySQL + NoSQL	Supabase Pro Managed PostgreSQL
Spatial DB	Proprietary / Custom	Not publicly disclosed	Custom Indexing	PostGIS (ST_DWithin spatial indexing)
Cache & Queue	Redis / Amazon ElastiCache	Not publicly disclosed	Redis	GCP Memorystore for Redis + BullMQ
Storage	AWS S3 / Internal	Not publicly disclosed	Cloud Storage	Supabase Storage + Cloudflare CDN
Container & Orchestration	Docker + Kubernetes	Not publicly disclosed	Docker + Kubernetes	Docker + GCP Cloud Run (asia-south1)
Cloud Infrastructure	AWS	AWS + Google Cloud	Google Cloud + AWS	Google Cloud Platform (GCP)
Security / Edge	Custom Edge Defense	Not publicly disclosed	Standard CDN	Cloudflare WAF + CDN + DNS + GCLB + Cloud NAT
Auth & Messaging	Custom OAuth / Phone Auth	Social / Phone Logins	Apple/Google SSO + Phone	Twilio WhatsApp OTP + Firebase Auth SMS Fallback + SSO
Real-time Comms	WebSockets, NATS	Chat, Group Chat, Audio Rooms	Chat, Video Calls, Real-time	Go Real-Time Engine + WebSockets
Audio / Video Calls	Integrated WebRTC	Agora / Custom VoIP	Agora / WebRTC	LiveKit Cloud (WebRTC SFU)
Payments	In-App Purchases (Stripe/App Stores)	Local Payment Gateways	App Store / Google Play / Stripe	Razorpay (UPI, Cards, Net Banking)
Notifications	FCM + APNs	Push Notifications	FCM + APNs	Firebase Cloud Messaging (FCM) + APNs
AI & Safety	Recommendation, Ranking, Safety	AI-powered safety & moderation	Recommendation, Matching, Safety	Google Cloud Vision (Vision API moderation & Safe Search) + Link Guard
Maps & Location	Google Maps API	Location Services	Google Maps API	MapLibre GL + Mapbox / OpenStreetMap
External Integrations	Spotify, Uber, etc.	Virtual Gifting, Games	Premium Subscriptions	LiveKit, Twilio, Razorpay, Firebase, Cloudflare
Key Takeaways	• Large-scale microservices • Kubernetes orchestration • Advanced ML matching • Dating focus	• Social interaction platform • Live audio rooms & games • AI moderation	• Native mobile apps • Matchmaking focus • Dating & friendship	• Single Flutter Codebase • GCP Cloud Run Tri-Microservices • LiveKit WebRTC & Twilio WhatsApp

3. DOMAIN MICROSERVICE MODULE INVENTORY

The platform decomposes domain capabilities across 11 discrete microservice modules running on the tri-service Cloud Run topology and asynchronous BullMQ worker runtime:

Module Name	Domain Capabilities, System Rules & Engineering Workflow	Runtime Service & SLA
1. Onboarding & Verification	9-step wizard, voice PIN recording, oval camera face liveness test. Twilio WhatsApp OTP verification with automated fallback to Firebase Auth Phone SMS. Google Cloud Vision profile photo pre-check via worker queue. Blue Tick auto-awarded at 70%+ completion.	NestJS API + Worker P0 (< 2s test)
2. Auth & Session Management	Dual-token model: 15-min access JWT + 30-day sliding refresh token rotation. Supabase Auth handles Google SSO and Apple SSO. Multi-device session revocation and token blacklisting stored in Memorystore Redis.	NestJS API P0 (< 100ms)
3. Home & Engagement	Coin wallet header, daily streak tracker, spin-wheel reward multiplier, personalized icebreaker cards, Top Picks feed, and dynamic profile completion nudges.	NestJS API P1 (Daily reset)
4. Discovery & Swipe Engine	Gesture deck (Like, Pass, Spark/Crush, Snooze, Revert). Server-enforced daily swipe limits. Hot-path candidate prefetching (10 cards) from Redis. PostGIS ST_DWithin geo-filtering and MapLibre GL map discovery.	Go Real-Time P0 (< 200ms)
5. Connect & Matchmaking	Mutual like detection, Spark/Mega-Crush high-priority push, 'Likes You' blurred profile stack, dynamic Vibe Match compatibility scoring, and reciprocal connection lifecycle.	Go Real-Time P1 (< 300ms)
6. Chats & Direct Messaging	1-on-1 direct messaging, message requests (text, Bitmoji, 2-min audio note), ephemeral vanish mode, typing indicators, read receipts, and in-flight link filtering.	Go Engine (WebSocket) P0 (< 500ms)
7. Call Engine & Media Sessions	LiveKit Cloud (Ship Plan) WebRTC peer-to-peer and SFU room routing. Random blind matching queue, camera preview safety scan, in-call mini-games, and server-authoritative 60-second heartbeat billing ticks.	Go Engine + LiveKit P2 (< 3s queue)
8. Profile & Media Storage	Verified Blue Tick badge, photo gallery supporting 6 profile photos + 1 appearance avatar per user, plus system reference strings for random anonymous call avatars stored in Supabase Storage with Cloudflare CDN, bio, and privacy controls.	NestJS API P0 (< 300ms)
9. Wallet & Monetization Ledger	Coin bundle purchase reconciliation, differential utility call rates, strict double-entry append-only transaction ledger, and verified creator cash-out processing (>500 coins).	NestJS API P1 (Idempotent)
10. Safety & Content Moderation	Asynchronous BullMQ background worker executing Google Cloud Vision API (Image Text Detection & Safe Search explicit content filtering), in-flight text regex/homoglyph contact guard, native OS screenshot capture alerts, and report queues.	Cloud Run Worker + Vision P0 (< 100ms)
11. Notifications & Telemetry	Asynchronous worker-dispatched FCM (Android) and APNs (iOS) push event fan-out for matches, likes, call invites, and streak reminders. Centralized logging and telemetry routed to GCP Cloud Logging & Cloud Monitoring.	Cloud Run Worker + Logging P1 (< 1s delivery)

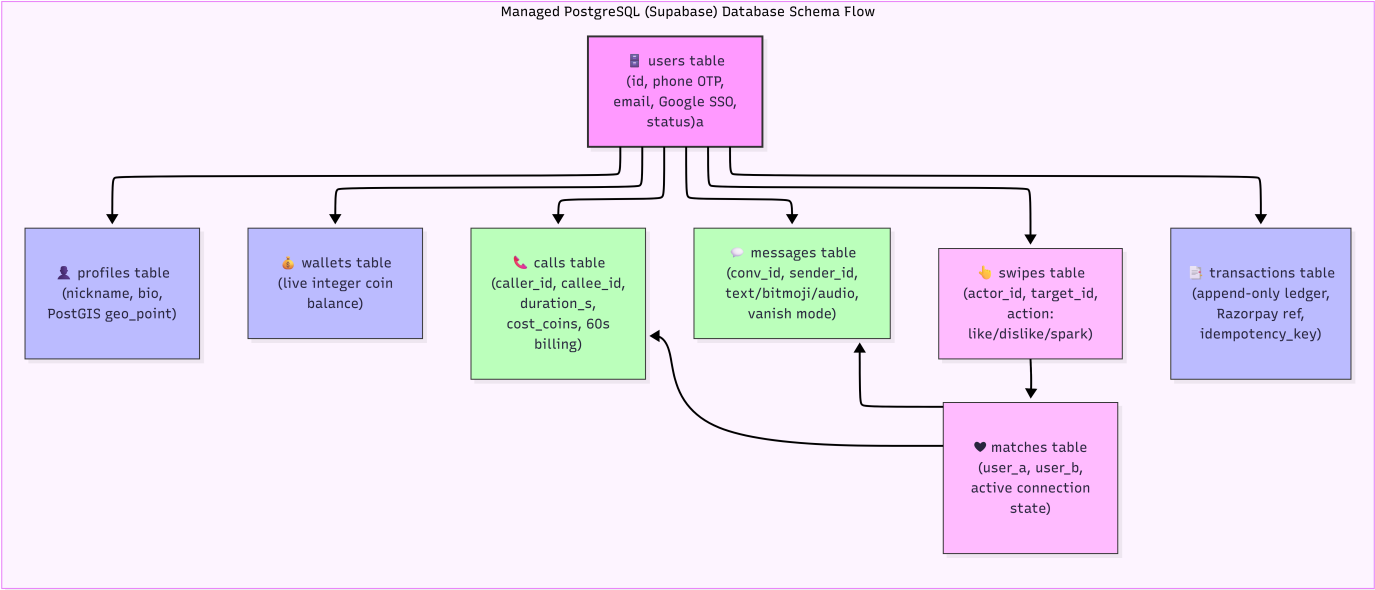
4. INFRASTRUCTURE & CONTAINER ORCHESTRATION (GCP CLOUD RUN & NETWORKING)

To properly isolate and scale the polyglot backend, YoUnMe implements a containerized microservices model deploying independent Docker containers onto Google Cloud Run in the Mumbai region. The architecture separates synchronous REST traffic (NestJS), persistent high-concurrency WebSockets (Go Engine), and asynchronous heavy workloads (Cloud Run Background Worker with BullMQ). This eliminates VM maintenance overhead, provides auto-scaling based on CPU/RAM and queue depth, and leverages Google Cloud's low-latency regional networking.

Subsystem	Cloud Service / Component	Operational Topology & Technical Configuration
Compute Services	Google Cloud Run (asia-south1 Mumbai)	Tri-service container deployment: NestJS API (0.25 vCPU, 512 MiB RAM, concurrency=20 req/inst, auto-scales 1-10); Go Engine (1.0 vCPU, 1.0 GiB RAM, concurrency=100 req/inst, auto-scales 1-20); Background Worker (1.0 vCPU, 512 MiB RAM, concurrency=10, auto-scales 0-5 based on BullMQ backlog).
Ingress Routing	Global Cloud Load Balancing (GCLB)	Single global forwarding rule acting as unified edge entry point. Directs /v1/* to NestJS API backend, /ws/* to Go WebSocket engine, and terminates TLS certificates with zero round-trip latency.
Secure Egress	Cloud NAT (Public NAT)	Public NAT configured with 2 VM instances and 1 dedicated static IP address. Provides secure, deterministic outbound traffic to external endpoints (Twilio WhatsApp, LiveKit Cloud, Firebase Auth/FCM, Apple APNs).
Perimeter Defense	Cloudflare Edge (Global Network)	Provides global authoritative DNS resolution, SSL/TLS edge termination, L3/L4/L7 DDoS mitigation, and edge caching for static assets, user avatars, and MapLibre map vector tiles.
Container Registry	GCP Artifact Registry (Asia Repository)	Regional container repository storing multi-stage Docker builds. Minimizes container pull latency and accelerates Cloud Run cold starts during horizontal autoscaling events.
Cache & Queue Engine	GCP Memorystore for Redis (Basic Tier, 1 GiB)	Managed Redis instance in asia-south1. Serves prefetch queues, active WebSocket session heartbeats, rate limiting, real-time application state, and BullMQ task queue management with sub-millisecond retrieval latency.

5. DATA ARCHITECTURE & SCHEMA SPECIFICATION

Figure 2: YoUnMe Managed PostgreSQL (Supabase Pro) Database Schema Flow Diagram



Core Relational Database Tables (Supabase Pro PostgreSQL + PostGIS)

Supabase Pro provides the primary database tier with 8 GB DB storage, 100 GB storage bucket, connection pooling, and PostGIS geo-queries for up to 100k MAU. Media storage supports 6 user photos + 1 appearance avatar per user, plus system reference strings for random anonymous call avatars:

Table Name	Primary / Foreign Keys	Core Column Definitions & Technical Constraints
users	id (PK, UUID)	phone (UNIQUE), email, google_id, dob, gender, is_verified, status (active/suspended), created_at, updated_at.
profiles	user_id (PK, FK users.id)	nickname, bio, city, languages (array), geo_point (PostGIS geometry(Point, 4326)), profile_progress (int), photo_urls (text array: 6 photos + 1 appearance avatar), call_avatar_ref (text).
swipes	id (PK, BIGSERIAL) actor_id, target_id (FKs)	action (like, dislike, spark), revert_flag (bool), snoozed_until (timestamptz), created_at. Compound index on (actor_id, target_id).
matches	id (PK, UUID) user_a, user_b (FKs)	matched_at (timestamptz), source (swipe, call), status (active, unmatched), UNIQUE(user_a, user_b).
messages	id (PK, BIGSERIAL) conv_id, sender_id (FKs)	type (text, bitmoji, audio), body (encrypted text), media_url, expires_at (vanish mode), is_flagged (bool), created_at.
calls	id (PK, UUID) caller_id, callee_id (FKs)	kind (audio, video, random), livekit_room_sid, started_at, duration_s, cost_coins, rating (1-5), status (completed, dropped).
wallets	user_id (PK, FK users.id)	balance (integer coins >= 0), updated_at. Managed via atomic decrement and server-authoritative database functions.
transactions	id (PK, UUID) user_id (FK users.id)	type (grant, purchase, spend, earn, payout), coins (integer), reference_id (text), idempotency_key (UNIQUE), created_at.

6. API & REAL-TIME EVENT ARCHITECTURE

Synchronous REST APIs and asynchronous WebSockets are divided across NestJS and Go Cloud Run services:

Domain	Endpoint & HTTP Method	Functional Scope, Authentication & Payload Protocol
Auth & Verification	POST /v1/auth/otp/whatsapp POST /v1/auth/otp/sms-fallback POST /v1/auth/otp/verify	Dispatches Twilio WhatsApp OTP template; triggers Firebase Auth Phone SMS fallback on failure. Verifies OTP and returns dual JWT payload.
Discovery	GET /v1/discovery/deck POST /v1/swipes GET /v1/discovery/map	Retrieves pre-cached candidate batch from Redis; records swipe action (like/reject/spark); provides MapLibre GL geo-tiles for map discovery.
Media Sessions	POST /v1/calls/token POST /v1/calls/{id}/heartbeat POST /v1/calls/{id}/end	Generates LiveKit Cloud WebRTC participant room token; processes 60-second billing heartbeat ticks; terminates call session and reconciles coins.
Wallet Ledger	GET /v1/wallet POST /v1/wallet/purchase POST /v1/wallet/payout	Fetches live coin balance and recent audit ledger entries; processes verified coin credit top-ups; triggers creator cash-out webhook.

Real-Time WebSocket Protocol (Go Engine):

- chat:{conv_id} – Low-latency 1-on-1 message streaming, typing indicators, read receipts, and vanish mode countdowns.
- presence:{user_id} – Real-time active status tracking backed by GCP Memorystore for Redis TTL keys.
- user:{user_id} – Push event pipeline for instant match popups, incoming Spark alerts, and coin balance updates.
- call:{call_id} – WebRTC session signaling, LiveKit room token exchange, and in-call mini-game synchronization.

7. MONETIZATION, COIN LEDGER & CREATOR ECONOMY ARCHITECTURE

The coin economy operates under strict server-authoritative rules. All coin allocations, spends, and transfers execute through a double-entry database ledger with unique idempotency keys, completely isolating business logic from external billing calculations:

Interaction Category	Male Account Action (Spend)	Female Creator Action (Earn)	Platform Verification & System Rules
Random Audio Call	20 coins / minute	5 coins / minute	Wallet pre-authorized; server-side 60s tick deduction via LiveKit webhook.
Random Video Call	50 coins / minute	10 coins / minute	Mandatory Google Cloud Vision preview scan; automated per-minute tick deduction.
Revert Swipe	5 coins / action (10 free/day)	N/A (System item)	Restores previous candidate card in Redis deck; daily quota resets midnight.
Spark / Mega-Crush	15 coins / action (20 free/day)	N/A (System item)	Bypasses candidate stack; delivers immediate high-priority push notification.
Direct Message Request	20 coins / request (15 free/day)	5 coins credited	Enforces 150-char text limit, Bitmoji gift, or 2-minute encrypted audio note.
Creator Cash-Out	N/A	Fixed token-to-payout conversion	Available strictly to verified profiles with >500 coins via secure payout webhook.

8. TRUST, SAFETY & IN-FLIGHT SECURITY PIPELINE

User trust, fraud prevention, and in-flight communication guardrails are enforced through automated microservice pipelines:

Security Trigger	Detection Mechanism & Integrated Service	Automated Enforcement & Mitigation Action
Trigger 1: Media Moderation	Google Cloud Vision API: - Image Text Detection (15k req/mo) - Safe Search explicit detection (30k req/mo)	Processed asynchronously via Cloud Run Background Worker (BullMQ). Scans profile photos and chat media for phone numbers, social handles, and explicit/racy content. Violative uploads are automatically blocked and flagged.
Trigger 2: Contact & Link Guard	Regex engine + homoglyph normalization scanning phone numbers, WhatsApp, Telegram, IG, and URLs.	Blocks client send button, displays link restriction toast, and logs strike count. Unlocks automatically after 3 days of matching or 20 message pairs.
Trigger 3: Screen Capture Guard	Native iOS/Android OS screenshot and screen recording broadcast events.	Displays instantaneous warning toast to offender, applies privacy overlay to media assets, and dispatches security alert to counterpart.

Contact Sharing Unlock Threshold: Server trust evaluation engine. Contact sharing restriction automatically unlocks after 3 days of active matching or 20 message pairs exchanged.

9. OBSERVABILITY & PHASED DELIVERY ROADMAP

Cloud Telemetry & Observability: Centralized monitoring is anchored on GCP Cloud Logging (50 GiB log ingestion, 1-month retention) and Cloud Monitoring. Alerting policies track Cloud Run CPU/memory limits, worker queue backlog via BullMQ, instance scaling thresholds, Memorystore Redis memory fragmentation, and LiveKit WebRTC packet loss metrics.

Phase	Technical Scope & Key Deliverables	Target Timeline	Production Exit Criteria
Phase 0: Foundation	Cloud Run tri-service setup in asia-south1, Artifact Registry CI/CD, Supabase Pro PostgreSQL + PostGIS baseline, Memorystore Redis + BullMQ queue setup.	2 - 3 Weeks	Twilio WhatsApp OTP, Firebase SMS fallback, and session rotation live on staging.
Phase 1: Core MVP	Onboarding wizard, Google Cloud Vision moderation via BullMQ worker, MapLibre GL discovery, swipe deck prefetch, 1-on-1 chat, safety link filter.	6 - 8 Weeks	Closed beta test matching, geo-filtering, and chat operational with zero security leaks.
Phase 2: Engagement & Queues	Home engagement feed, Connect tabs, message requests, double-entry wallet ledger, daily streak engine, BullMQ batch FCM/APNs push pipeline.	5 - 6 Weeks	Idempotent coin ledger verified; daily streak retention uplift confirmed.
Phase 3: Media & Scale	LiveKit Cloud WebRTC integration, random blind call queue, 60s billing heartbeat ticks, verified creator cash-out module.	5 - 6 Weeks	WebRTC call setup latency < 2s; automated creator cash-out reconciliation verified.